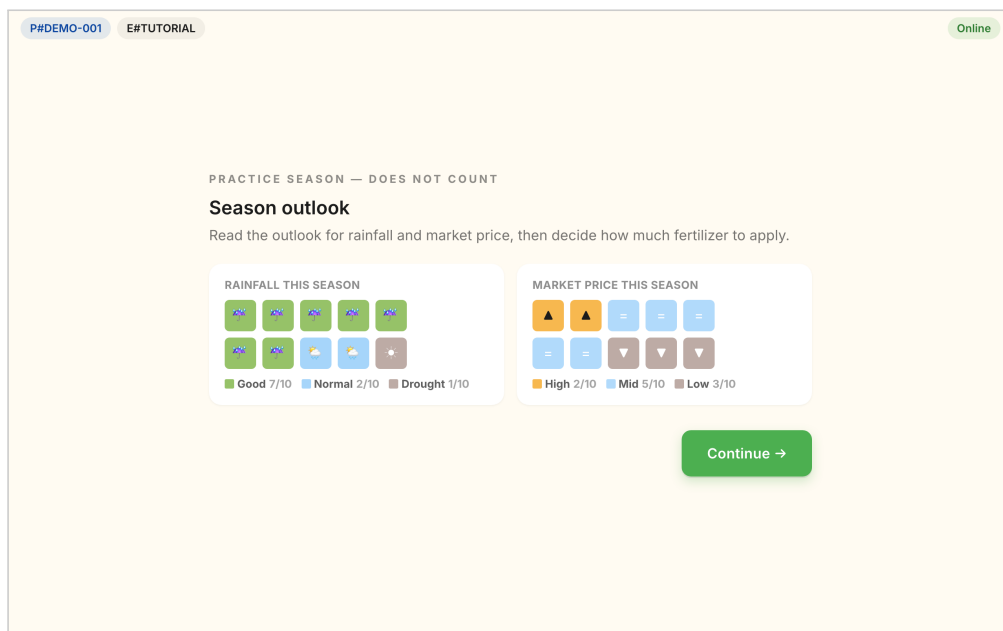


The Fertilizer Game

A Step-by-Step Guide for Farmers

How to play, and what it teaches about using fertilizer
when the rain and the market price are uncertain



CIMMYT Field Experiment Tool

Maize · Nigeria · English / Hausa

This is a practice game played on a tablet. The money is play-money (*tokens*); at the end your tokens are changed into real Naira.

1. What is this game about?

In this game **you are a maize farmer**. Each **season** you must make one decision: **how much fertilizer to buy for your field**.

The catch is that two things are **uncertain**, just like in real farming:

- **The rain.** It can be **good**, **normal**, or there can be a **drought**. Fertilizer only helps your maize grow when there is enough water. In a drought, fertilizer does almost nothing.
- **The market price.** When you sell your maize the price can be **high**, **normal (mid)**, or **low**. A good harvest is worth less money when the price is low.

Before each season the game *tells you the chances* of good rain, drought, high price, and so on. Your job is to look at those chances and decide how much fertilizer is worth buying.

The one thing to remember

Fertilizer is like an **investment**: you spend some of your money on it, hoping the extra harvest will be worth *more* than what you spent. Sometimes it pays off well. Sometimes — in a drought, or when the price is low — it does not, and you **lose** the money you spent on it. Buying *no* fertilizer is always safe: you simply keep your money.

2. What you start with each season

At the start of **every** season you are given **25 tokens** (think of tokens as money).

- Each **bag of fertilizer** costs **1 token**.
- You can buy from **0** bags (none) up to **10** bags.
- Any tokens you do **not** spend, you **keep**.

So if you buy 4 bags, you spend 4 tokens and keep 21. Those 4 bags are your **investment** — their job is to grow extra maize that you can sell.

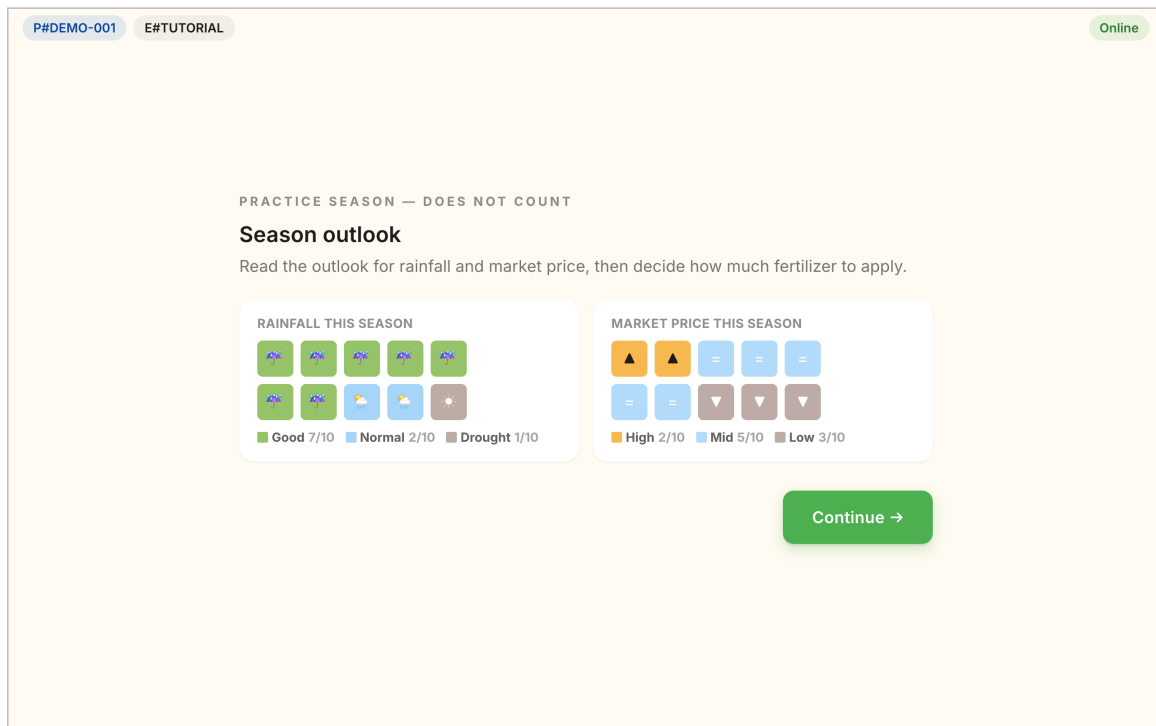
There is **1 practice season** first (it does not count), and then **10 real seasons**. Your tokens from the 10 real seasons are added up and changed into real money at the end.

3. One season, step by step

Every season has the same five steps.

Step 1 — See the outlook (the chances)

First you see the **chances** for this season: one picture for the **rain** and one for the **market price**. Each picture has **10 small squares** that stand for 10 possible seasons.



The season outlook. **Left:** rainfall — here 7 of 10 squares are *Good* (green), 2 *Normal* (blue), 1 *Drought* (grey). **Right:** market price — 2 of 10 *High*, 5 *Mid*, 3 *Low*.

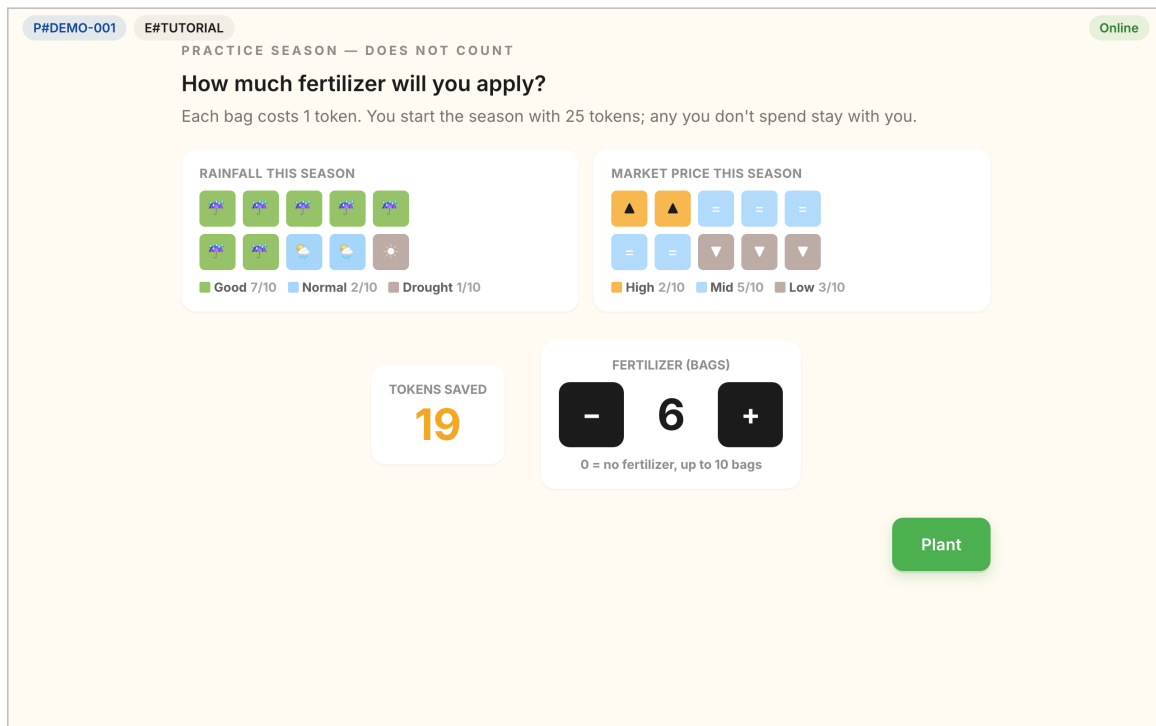
How to read it: in the example above, **7 out of 10** squares show good rain, so good rain is *very likely* this season. Only **1 out of 10** shows drought, so a drought is *unlikely*. The more green squares, the safer it is to buy fertilizer.

Reading the pictures

- **Rain — Good** the best case; fertilizer works very well
- **Rain — Normal** fertilizer works, but less than in good rain
- **Rain — Drought** fertilizer does **nothing**; you lose what you spent
- ▲ **Price — High** your maize sells for the most
- = **Price — Mid** a normal price
- ▼ **Price — Low** your maize sells for little

Step 2 — Decide how much fertilizer

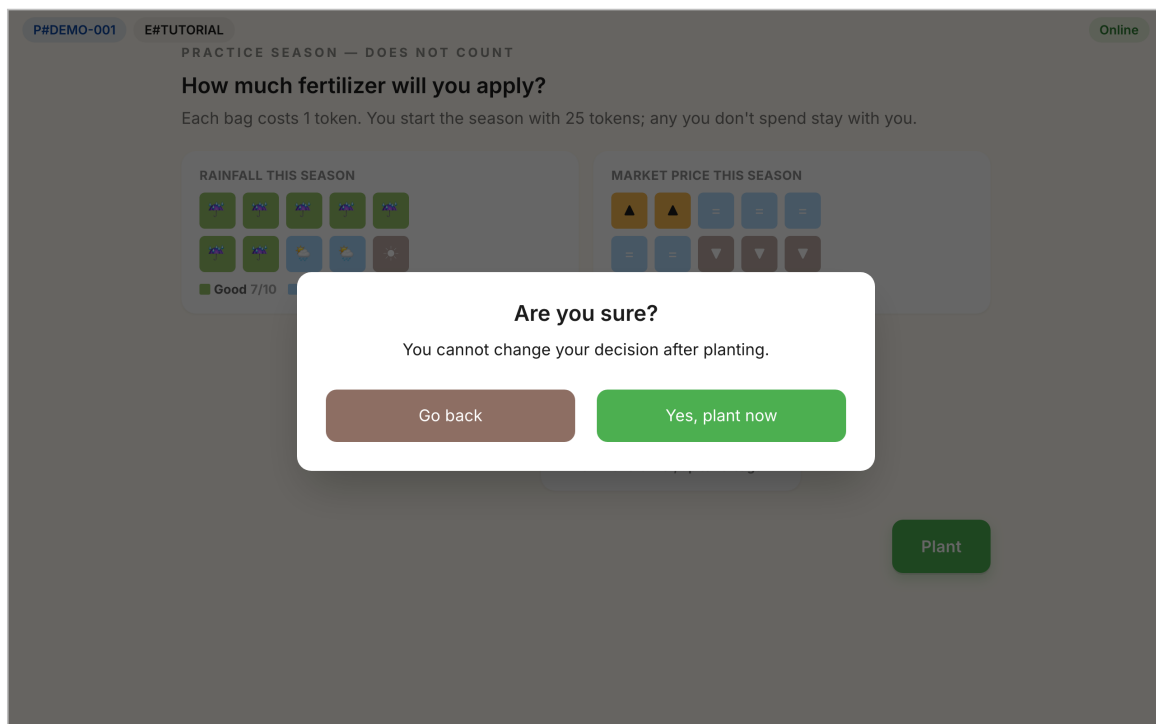
Now you choose your number of bags with the **minus** (–) and **plus** (+) buttons. As you change it, you can see how many tokens you are keeping (“Tokens saved”). The rain and price chances stay on the screen so you can keep looking at them while you decide.



*Choosing the dose. Here 6 bags are selected. The card on the left shows the tokens you would keep. Tap **Plant** when you are happy with your choice.*

Step 3 — Confirm (you cannot change it later)

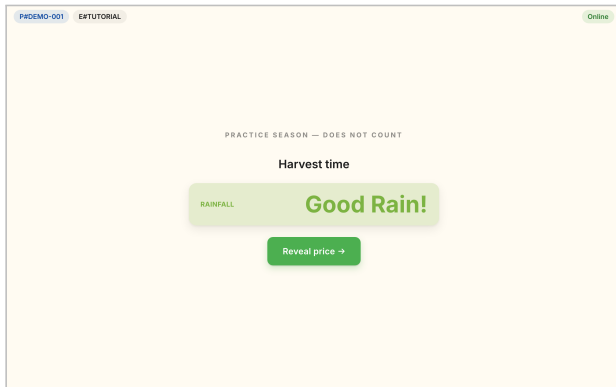
Once you tap Plant, the game asks you to confirm. **After you confirm you cannot change your decision** — just like in real life, once you have planted, you cannot un-plant.



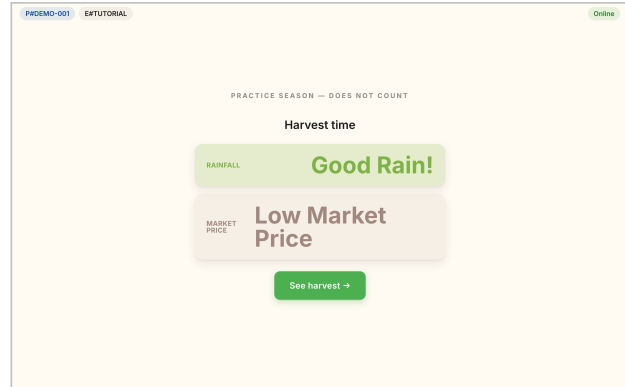
*The confirmation. Tap **Go back** to change your mind, or **Yes, plant now** to lock in your decision.*

Step 4 — See what happened

Now the season plays out. First the **rain** result is shown, then the **market price**. This is decided by chance, using the odds you saw in Step 1 — nobody can change it.



First: the rain result.

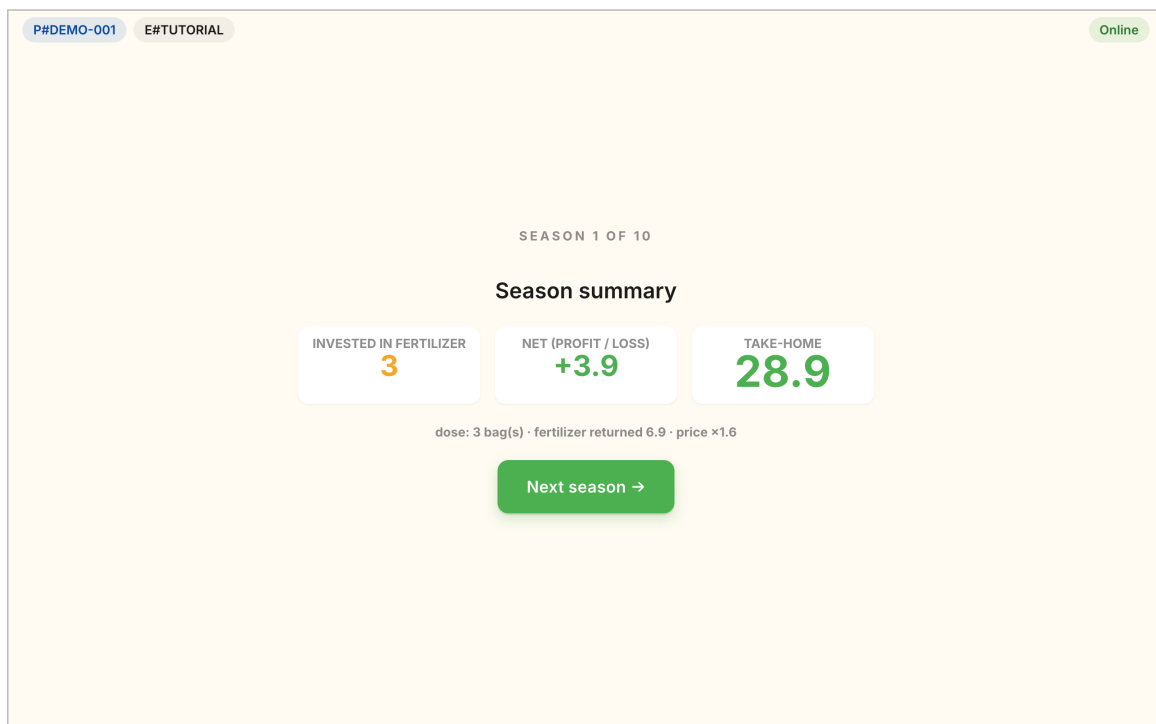


Then: the market price.

Step 5 — Your season result

Finally you see the result of your decision, shown in three simple numbers:

- **Invested in fertilizer** — the tokens you spent on bags.
- **Net (profit / loss)** — did the extra harvest earn back *more* than you spent (**green, a profit**) or *less* (**brown, a loss**)?
- **Take-home** — the tokens you actually keep this season.



*A good season. The farmer invested 3 bags. The rain was good and the price was high ($\times 1.6$), so the fertilizer returned 6.9 — much more than the 3 spent. **Net** = +3.9, and take-home is 28.9 tokens.*

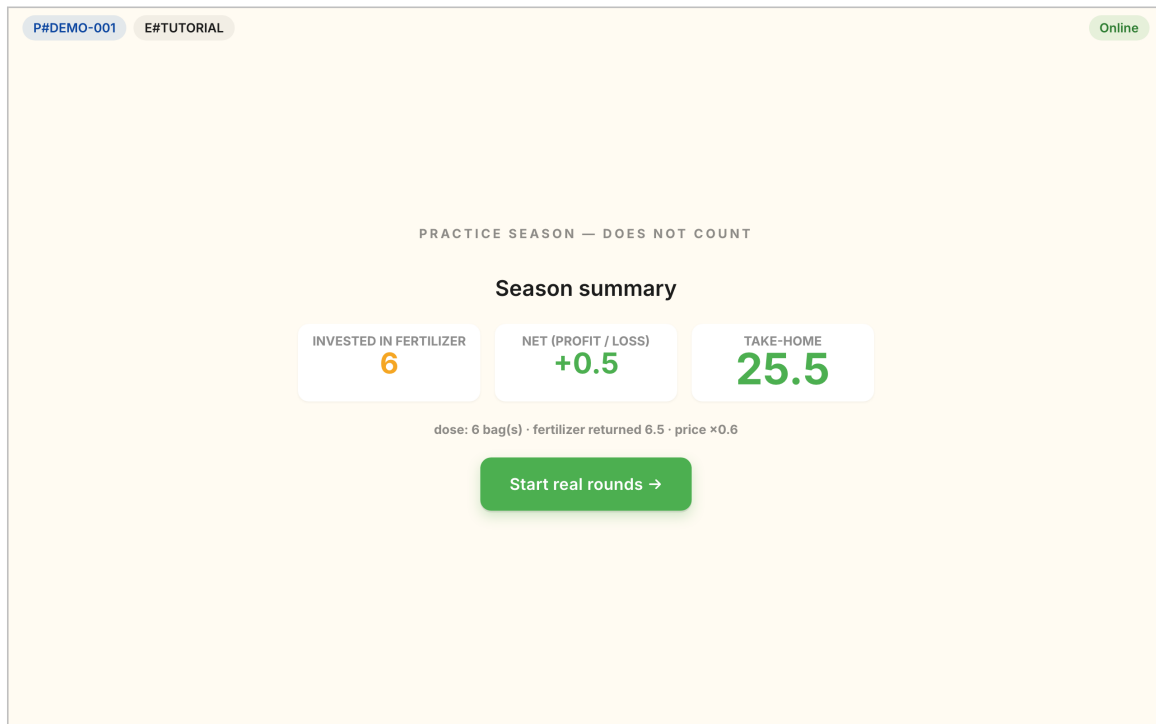
4. The big lesson: fertilizer can win *or* lose

The same number of bags can give very different results, depending on the rain and the price. Below are three real season results from the game.

A win — good rain and high price

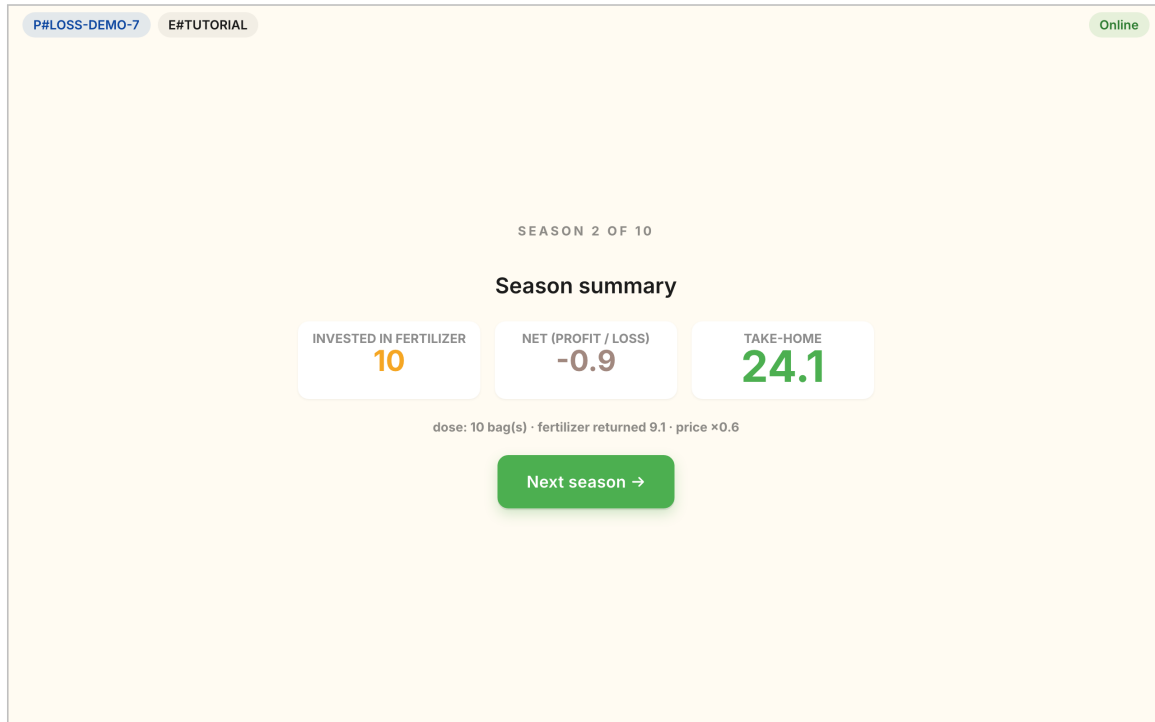
Already shown above: **invest 3, return 6.9, net +3.9**. When the weather and the price are both good, fertilizer pays off well.

Barely worth it — good rain but low price



Here the farmer invested **6** bags. The harvest grew, but the market price was **low** ($\times 0.6$), so the fertilizer only returned **6.5**. **Net** = **+0.5** — almost exactly break-even. A low price can wipe out your profit.

A loss — the price did not cover the cost



Here the farmer invested **10** bags, but the price was **low** ($\times 0.6$) and the fertilizer only returned **9.1** — less than the 10 spent. **Net** = **-0.9**: a loss. **Take-home** (24.1) is below the 25 they started with.

What a drought does

The worst case is a **drought**. In a drought, fertilizer does **nothing** extra — so the return is **0**, and you **lose every token** you spent on bags. For example, if you buy 8 bags and a drought comes, your return is 0 and your net is -8 . This is why, when the outlook shows many grey (drought) squares, buying a lot of fertilizer is risky.

The four cases, in plain words

Good rain & high price	fertilizer pays off well — a clear profit
Good rain & low price	the harvest is good but sells cheap — small profit or break-even
Drought	fertilizer does nothing — you lose what you spent
Buy no fertilizer	you spend nothing and keep all your money — always safe

5. How to decide each season

There is no single “correct” number of bags — it depends on the chances you are shown. A good way to think:

1. **Look at the rain picture first.** Many green (good) squares and few grey (drought) squares? Then fertilizer is more likely to pay off, and buying more makes sense.
2. **Then look at the price picture.** Many high/mid squares means your harvest will sell well. Many low squares means even a good harvest earns little — be more careful.
3. **When in doubt, buy less.** Buying fewer bags means you risk fewer tokens. Buying *no* fertilizer is never a loss.

4. **There is no debt.** Even in the worst season you can never go below the tokens you did not spend — you will always take *some* money home.

6. The whole game, start to finish

1. A short lesson with pictures, to make sure the rain and price pictures are clear.
2. **1 practice season** — to try the steps. It does **not** count.
3. **10 real seasons** — each one is the five steps above. Your take-home from each is saved.
4. **Final result** — your tokens from all 10 seasons are added up and changed into real Naira.

Remember

Play each season as if the tokens were your own money — because at the end, they become real money. Read the pictures, think about the chances, and choose the amount of fertilizer *you* think is worth the risk.